Managing virtual machines

vmware_guest: This module is required in order to manage a virtual machine in your vCenter or standalone ESXi environment. The module can create, modify, rename and remove a virtual machine. It can also power on, power off, restart and suspend a virtual machine.

The following Ansible Playbook example shows how to create a virtual machine:

```
- name: Example Playbook for creating virtual machine
  hosts: localhost
  tasks:
    - name: Create a virtual machine with name testvm01
      vmware_guest:
        datacenter: datacenter01
        esxi_hostname: esxi_host_01
        hostname: 127.0.0.1
        name: testvm01
        guest_id: centos64guest
        username: user
        password: pass@123
        state: present
        disk:
          - size_gb: 1
            type: thin
            datastore: LocalDS_0
        hardware:
            memory_mb: 512
            num_cpus: 1
            osid: centos64guest
            scsi: paravirtual
```

You can read more about this module at *http://docs.ansible.com/ansible/latest/vmware_guest_module.html*.

vmware_guest_facts: You often want to get details about a virtual machine like its IP address, folder path, MAC address, etc. This module does exactly that. *vmware_guest_facts* gives you all the vital information about a virtual machine. You can chain the information received from this module to various other VMware modules.

```
---
- name: Example playbook to get facts about virtual machine
  hosts: localhost
  tasks:
    - name: get list of facts about virtual machines
      vmware_guest_facts:
        validate_certs: False
        hostname: esxi_hostname_01
        username: user
        password: pass@123
        datacenter: datacenter_01
        name: testvm01
        folder: /vm
      register: guest_facts_0001
    - debug: msg="{{ guest_facts_0001 }}"
```

You can read more about this module at *http://docs.ansible.com/ansible/latest/vmware_guest_facts_module.html*.

vmware_guest_find: This module provides information about the *folder path* in which the virtual machine is located. This information is required in various other VMware modules.

The following example will find the *folder path* related to a virtual machine.

```
---
- name: Example playbook to find the details of folder related
to virtual machine
  hosts: localhost
  tasks:
    - name: Get folder path for virtual machine
      vmware_guest_find:
        validate_certs: false
        hostname: 127.0.0.1
        username: user
        password: pass
        name: testvm01
        datacenter: datacenter_01
      register: folder_details
- debug: msg="{{ folder_details }}"
```

You can read more about this module at *https://github.com/ansible/ansible/blob/devel/lib/ansible/modules/cloud/vmware/vmware_guest_find.py*.

vmware_guest_snapshot: When we want to preserve the state of the virtual machine at a given point of time, we take a snapshot of the virtual machine. This module provides the functionality to manage various operations related to a snapshot such as *create*, *delete*, *revert* and *restore*. This module requires information on the *folder path* along with the virtual machine name. Here, we can use the information provided by *vmware_guest_find* on the *folder path*.

The following playbook example will show how to take a simple snapshot of a virtual machine:

```
---
- name: Example to take a snapshot of virtual machine
  hosts: localhost
  tasks:
    - name: Take a snapshot
      vmware_guest_snapshot:
        hostname: esxi_hostname_01
        username: root
        password: esxi@123
        name: testvm01
        folder: /vm
        snapshot_name: my_secure_stable_snapshot
      register: snap_state_001
    - debug: msg="{{ snap_state_001 }}"
```